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Mattiq respectfully submits this comment to the Office of Science and Technology Policy in support of the development of America's AI Action Plan. The rapid innovation of AI seen thus far—and the need for America to maintain and extend its dominance and leadership in AI innovation and development—is of paramount importance to Mattiq. We are grateful that "it is the policy of the United States to sustain and enhance America's global AI dominance in order to promote human flourishing, economic competitiveness, and national security,"[1] and for the opportunity to contribute to the development of the AI Action Plan.

We are an Illinois-based startup spun out of Northwestern University, building the world's most technologically advanced materials and chemistry discovery & design company. Mattiq was founded in 2021 by Dr. Chad A. Mirkin, the George B. Rathmann Professor of Chemistry and Director of the International Institute for Nanotechnology at Northwestern University, and Dr. Andrey Ivankin, the CTO, who leads the development of the Mattiq's core ultrahigh-throughput experimentation, data, and AI technology.

We are pioneering revolutionary ultrahigh-throughput technologies for accelerating the discovery and development of novel materials and the breakthrough applications they enable, while generating the largest, highest-quality, and most diverse datasets to train the most advanced materials AI models. Not only can these technologies allow America to outpace global competitors in developing superior materials and breakthrough technologies across key industries, but they can also strengthen supply chain resilience for critical material technologies—reducing reliance on scarce strategic and "rare earth" minerals essential for America's defense and economy.[2] Moreover, these technologies will

[1] *Removing Barriers to American Leadership in Artificial Intelligence*, Exec. Order No. 14,179, 90 Fed. Reg. 8,741 (Jan. 31, 2025).

[2] *See, e.g.*, *Unleashing American Energy*, Exec. Order No. 14,154, 90 Fed. Reg. 8,353 (Jan. 29, 2025) ("**Unleashing American Energy**") at § 2(b)-(c) (stating it is the policy of the United States "to establish our position as the leading producer and processor of non-fuel minerals, including rare earth minerals, which will create jobs and prosperity at home, strengthen supply chains for the United States and its allies, and reduce the global influence of malign and adversarial states" and "to protect the United States's economic and national security and military preparedness by ensuring that an abundant supply of reliable energy is readily accessible in every State and territory of the Nation."); *see also Declaring a National Energy Emergency*, Exec. Order No. 14,156, 90 Fed. Reg. 8,433 (Jan. 29, 2025) ("**National**

cement America's leadership in the global materials AI race, securing long-term technological and economic dominance.

However, much like with the race for Artificial General Intelligence, the race to achieve a true Artificial General Materials and Chemistry Intelligence—which will enable the winner to seize strategic military and economic dominance in countless areas—will be a long and resource-intensive journey. In our comment, we explain the strategic importance of AI for materials innovation, why the "data gap" for materials discoveries is the bottleneck preventing America from fully harnessing AI to discover advanced strategic materials, how Mattiq is working to close that gap to provide America with the opportunity to fully unleash AI's potential for materials discoveries, and our recommendations on the AI Action Plan for America to win that race.

Strategic Importance of AI for Materials Innovation

Geopolitical and economic leadership has long been driven by breakthroughs in materials science and the transformative technologies they enable. From the silicon that fueled the semiconductor revolution to advanced alloys that power quantum computing and form the backbone of modern aerospace and defense, materials innovation remains a cornerstone of national power. History has shown that breakthroughs in materials can reshape global power dynamics. Just as the silicon transistor cemented U.S. technological dominance for decades, China's near-monopoly on rare earth elements in the early 2010s illustrates how control over critical materials can be leveraged for economic and strategic advantage. Sustaining leadership in this domain is not merely advantageous; it is a strategic imperative for technological supremacy, economic resilience, and national security.

AI has the potential to revolutionize materials discovery by dramatically reducing timelines from years to mere months or weeks while significantly lowering costs. Leading the race in AI for materials discovery is of immense importance, as it will allow the U.S. to outpace global competitors in developing superior materials and breakthrough technologies across key industries and strengthen resilience of materials supply chain for critical technologies.

Energy Emergency") at § 1 ("The energy and critical minerals ("energy") identification, leasing, development, production, transportation, refining, and generation capacity of the United States are all far too inadequate to meet our Nation's needs. We need a reliable, diversified, and affordable supply of energy to drive our Nation's manufacturing, transportation, agriculture, and defense industries, and to sustain the basics of modern life and military preparedness"; "The United States' insufficient energy production, transportation, refining, and generation constitutes an unusual and extraordinary threat to our Nation's economy, national security, and foreign policy.").

Materials Data: the Biggest Challenge and the Greatest Opportunity

Despite AI's impressive current capabilities, a "data gap" prevents full harnessing of AI's potential to rapidly make advanced materials discovery. America must close that gap before its adversaries do so.

Modern AI algorithms require vast, high-quality training datasets to achieve "superhuman" performance—yet materials datasets remain scarce, fragmented, and inconsistent. Without sufficient examples to learn the underlying chemistry and physics, current AI models—including the major large language models (LLMs) like ChatGPT, Claude, and Gemini—struggle to make accurate predictions and generalize beyond known materials. That is because training models on the same generally available datasets causes their performance to converge, highlighting that true differentiation comes from proprietary and highly reliable datasets.

But the design universe for materials—their composition and structure, properties, and environmental interactions—is extraordinary diverse and nearly infinite, posing a significant challenge in generating adequate training datasets and developing transformative AI models that can generalize across a wide range of materials problems. Moreover, computational modeling data frequently lack experimental validation, while experimental data are still collected in a near-linear, slow manner, stored in siloed databases, and recorded in inconsistent machine-unreadable formats. Additionally, traditional methods can effectively explore and generate data for only a tiny fraction—less than one percent—of the vast materials design universe, leaving the opportunity to discover significantly superior materials largely out of reach.

In sum, while advancements in materials AI algorithms are crucial for winning the race to fully harness AI for materials discoveries, the true key to unlocking AI's full potential today lies in closing the data gap: whoever achieves this first will secure a lasting competitive advantage in the next era of materials innovation.

Mattiq's Unparalleled Materials Data-Generation Engine: Fueling AI to Exponentially Accelerate the Discovery of Advanced Materials Critical to America's Defense and Economy

Mattiq is building the fastest, most scalable, and generalizable materials data-generation engine in the industry. Our unrivaled ultrahigh-throughput synthesis technology enables access to breakthrough materials from a vastly expanded design space, accelerating innovation across industries—including chemicals, catalysis, photonics, optoelectronics, electronics, semiconductors, aerospace, defense, and energy—at speeds thousands to millions of times faster than the next best-in-class approach. By developing complementary high-throughput screening and end-use validation capabilities in an application-specific manner, we are creating vast and continuously expanding training datasets essential for building transformative AI models that can generalize across a wide range of materials problems.

Mattiq's core proprietary "Megalibrary" technology—enabled by breakthroughs in nanotechnology, chemistry, and engineering—revolutionizes materials science and chemistry in a way akin to how genomic chip technology transformed bioscience and drug discovery. Megalibrary two-by-two centimeter chips enable us to positionally encode over 100 million unique nanomaterial candidates, each with complex yet precisely controlled composition, size, and structure, unlocking previously unimaginable throughput and design space for materials explorations and data generation.

Similar to drug discovery and development, high-throughput discovery with Megalibraries does not always guarantee successful translation into end-use applications but still provides a vastly superior starting point for today's materials development workflow. By scaling up, integrating into end-use systems, and validating the performance of Megalibrary discoveries—either in-house or through strategic partnerships—Mattiq continuously refines its Megalibrary technology, maximizing the relevance of generated data and increasing the likelihood of successful deployment.

Crucially, all Megalibraries materials data are generated under highly unified and controlled experimental parameters, processed through standardized quality assurance pipelines, and securely stored in a secure database: ensuring consistency, reliability, and scalability. Not only will these datasets enable the development of state-of-the-art materials AI models, but they will also serve as the ground truth for theoretical frameworks and computational modeling, allowing the creation of more accurate models for complex nanoscale materials and processes. Ultimately, this will pave the way for the purely synthetic design of novel materials with desired unprecedented properties, eliminating the need for physical experiments.

Recommendations for America's AI Action Plan

Mattiq believes that, among other important priorities, America's AI Action Plan should emphasize the following points.

Infrastructure. America needs to make national investments to establish cutting-edge shared advanced materials characterization hubs, equipped with the most sophisticated materials and chemistry characterization tools. Additionally, America should develop hubs dedicated to scaled-up synthesis, end-use system integration, and validation of new materials to accelerate real-world deployment.

America should also create incentives and funding mechanisms for instrument manufacturers to drive advancements in massively parallel and high-throughput experimentation and automation, ensuring the infrastructure keeps pace with the rapid evolution of materials discovery.

America should also expand access to computational resources, making high-performance computing more abundant and widely available to researchers and innovators. These critical resources are beyond the reach of most groundbreaking startups, yet they are essential for unlocking AI's full potential in materials science and maintaining U.S. leadership in the global materials innovation race.

Talent. America needs to develop a talent pipeline that bridges AI and materials science, ensuring that America has the human capital to sustain leadership in this important field. We recommend new education and training initiatives at all levels. Universities should be supported (through NSF grants, for example) to create interdisciplinary curricula that blend materials science, computer science, and applied math, producing graduates fluent in all of those domains. Federal fellowships or traineeships can incentivize Ph.D. students to pursue cross-disciplinary research in AI-guided materials design. Likewise, national labs, and industry could offer internship programs for young scientists and engineers to gain hands-on experience with AI-driven experimentation platforms.

Investment. While of immense national importance, developing this capability is highly resource-intensive, and the gap between innovation and commercial success—the "valley of death"—is simply too vast to be bridged by private investment alone. Strategic public support is essential to ensure these groundbreaking advances reach real-world applications and secure U.S. leadership in materials innovation.

Industrial Collaboration through Policy. It is also essential for America to foster a closer interface between high-throughput experimentation companies, AI companies and industrial players to accelerate the real-world impact of AI-driven materials discovery. America should take a whole-of-nation approach and incentivize companies to pursue the development of novel materials—enhancing their competitiveness and strengthening supply chain resilience, particularly for nationally critical technologies—to drive deeper industry engagement. Additionally, providing a clear framework and guidance for intellectual property in the context of data sharing and AI model development can help break down silos, encourage collaboration, and accelerate the commercialization of AI-driven materials innovations.

National Interest Policy. Finally, America should also provide strategic mechanisms and incentives to encourage companies developing materials data at scale and advanced materials AI to prioritize investments and partnerships within America. Ensuring that these critical assets remain within an America-centered innovation ecosystem will help safeguard economic and technological leadership in AI-driven materials discovery.

These proposals should accelerate America's dominance and increase its chance of winning the race to achieve AI-enabled Artificial General Materials and Chemistry Intelligence. And Mattiq is an example of how some of the existing investments and partnerships between the government and private industry have already shown promise. Mattiq is the grateful recipient of grants designed to advance America's leadership in cutting-edge technologies, including grants from the NSF's Small Business Innovator Research (also known as America's Seed Fund powered by NSF) ("**SBIR**") Phase I program,[3] as well as the U.S. Department of Energy's ("**DOE**") Advanced Research Projects Agency-Energy program ("**ARPA-E**"), High-Performance Computing for Energy Innovation program ("**HPC4EI**").[4]

Despite the somewhat modest sums we have received in grants awarded to date (in an aggregate amount of about $2.6 million), we have already developed, in the four short years since our founding, a portfolio of novel, highly durable alternatives to strategically critical element: iridium.[5] Iridium is a rare and expensive element, and is considered to be "critical mineral" by DOE,[6] as well as a "strategic material of interest" by the U.S. Defense Logistics Agency due to its use in guided missile systems, military semiconductors, and aircraft engines.[7] Iridium is also critical to the production of diversified energy sources because it is a catalyst for the oxygen evolution reaction that is used to make hydrogen from water, making it important to America's goal to further unleash the production of reliable energy domestically.[8] However, iridium is both rare and expensive, and is primarily sourced from South Africa and Russia, posing geopolitical challenges.[9] Speeding the discovery of new materials to replace iridium—as well as other critical and rare elements critical to America's defense and economy—requires intentional and intense increases in funding from federal agencies to companies, like ours, committed to putting America first. While iridium substitutes are one initial area of focus for Mattiq, countless other areas exist for similar innovation in the interests of resource diversity, enhanced performance, and raw discovery.

Implementing our proposals above, some of which involve increasing America's existing investments and commitments to partnerships with private industry, will simultaneously advance several paramount priorities of America, such as increasing America's dominance and leadership in AI technologies, freedom from existing imbalances in access to

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[3] *See*, https://www.nsf.gov/awardsearch/showAward?AWD_ID=2420066, https://www.nsf.gov/awardsearch/showAward?AWD_ID=2151713.

[4] *See* https://hpc4energyinnovation.llnl.gov/projects.

[5] *See* https://www.businesswire.com/news/home/20231003314654/en/Mattiq-Develops-Portfolio-of-Novel-Catalysts-to-Solve-Key-Materials-Challenge-in-the-Global-Scale-Up-of-Clean-Hydrogen.

[6] *See Notice of Final Determination on 2023 DOE Critical Materials List*, 88 Fed. Reg. 51,792 (Aug. 4, 2023) ("**DOE Critical Materials List**").

[7] *See* https://www.dla.mil/Strategic-Materials/Materials/.

[8] *See* Unleashing American Energy at § 2(c); *see also* DOE Critical Materials List.

[9] *See, e.g.*, *Addressing Egregious Actions of the Republic of South Africa*, Exec. Order No. 14,204, 90 Fed. Reg. 9,497 (Feb. 12, 2025) at § 1 ("In addition, South Affrica has taken aggressive positions towards the United States and its allies, . . .").

rare earth or alternative materials, and the further development of state-of-the-art weapons and defense systems.[10]

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In closing, Mattiq is an enthusiastic supporter of America's efforts to take the opportunities presented by AI and other advanced technologies seriously. We stand ready to help advance and achieve America's priorities and look forward to further collaboration on our important work.

Very truly yours,

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Dr. Andrey Ivankin
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Mattiq, Inc.

[10] *See, e.g.*, *Establishing the National Energy Dominance Council*, Exec. Order No. 14,213, 90 Fed. Reg. 9,945 (Feb. 20, 2025) at § 1 ("We must expand all forms of reliable and affordable energy production to drive down inflation, grow our economy, create good-paying jobs, reestablish American leadership in manufacturing, lead the world in artificial intelligence, and restore peace through strength by wielding our commercial and diplomatic levers to end wars across the world. By utilizing our amazing national assets, including our crude oil, natural gas, lease condensates, natural gas liquids, refined petroleum products, uranium, coal, biofuels, geothermal heat, the kinetic movement of flowing water, and critical minerals, we will preserve and protect our most beautiful places, reduce our dependency on foreign imports, and grow our economy—thereby enabling the reduction of our deficits and our debt.").